

# Blind Disturbance-Model Selection for Diffusion-Aided Semantic Image Reception

Author information pending

Affiliations and correspondence to be supplied

**Abstract**—Explicit interference reconstruction is useful only when its additional modeling assumptions improve the desired image estimate. With unknown interference power, a joint diffusion sampler may instead introduce errors and unnecessary computation. We propose a blind disturbance-model selector for a frozen semantic image codec. Before diffusion inference, the receiver uses excess received energy and block-energy heterogeneity to choose either Gaussian denoising or joint signal–interference reconstruction. The rule is fitted on separate development images; only the selected branch is executed, and no neural weights are updated. We evaluate the rule on CelebA images with CIFAR-10 interference, using identical received signals for all receiver comparisons and a joint receiver with the same bypass as an additional control. On 256 independent image pairs with two random seeds and 18 channel conditions, the frozen rule attains 21.593 dB mean PSNR: +1.334 dB relative to calibrated joint sampling and +0.750 dB relative to the blind Gaussian receiver. It uses 52.2 total predictor evaluations per frame on average, compared with 80 for joint sampling. Separately trained DeepJSCC and MambaJSCC codecs provide external performance references under the same channel-use budget.

**Index Terms**—Semantic communications, interference cancellation, diffusion models, receiver adaptation, deep joint source–channel coding.

## I. INTRODUCTION

Deep joint source–channel coding (JSCC) maps an image directly to channel symbols, allowing the decoder to learn robustness to transmission noise [1], [2]. A deployed codec can nevertheless encounter co-channel interference absent from its training channel. Diffusion priors offer two ways to process the received symbols. Channel denoising diffusion models (CDDM) motivate treating the disturbance as Gaussian [3], whereas interference cancellation diffusion models (ICDM) represent the desired and interfering signals explicitly [4].

The more detailed model is not always the better receiver. Gaussian denoising ignores interference structure but avoids estimating a second latent signal. Joint reconstruction uses that structure, at the cost of estimating both the interferer and its unknown amplitude from one mixture. Errors in those estimates can degrade the desired image even when the reconstructed mixture fits the observation. A fixed choice therefore risks either discarding useful structure or paying for unreliable separation.

We address the resulting decision: *when should a frozen receiver denoise the aggregate disturbance, and when should it reconstruct the interferer?* The decision must precede diffusion inference. Running both receivers and choosing the better reconstruction would require an unavailable quality reference

and incur both inference costs. Our selector instead computes two statistics from the received symbols: excess energy estimates disturbance strength, while block-energy heterogeneity describes variation across symbol blocks. A threshold rule learned on separate development images selects one branch. The codec and both diffusion priors remain frozen.

The contribution has three parts:

- a blind selection rule that chooses the disturbance model before sampling, using  $O(n)$  received-energy calculations and no additional neural inference;
- a development-only fitting procedure that compares energy-only and two-feature rules, and retains a fixed receiver when validation finds no gain;
- an independent paired evaluation that separates model selection from power estimation and direct-decoding bypass, and measures both reconstruction gains and harmful decisions.

The key comparison is against a Gaussian receiver calibrated to *total* disturbance, not thermal noise alone. A joint receiver with the same bypass provides a second control for improvements caused merely by skipping separation. These controls test the receiver decision without changing the underlying learned representations.

## II. RELATED WORK

CDDM matches channel noise to a diffusion noise level and jointly considers codec training, channel denoising and decoder adaptation [3]. ICDM adds an interference prior for joint signal recovery [4]. We use these two receiver principles with shared frozen weights to study their selection. The resulting Gaussian and joint receivers are receiver-only ports; they do not reproduce the complete published training pipelines. DeepJSCC [1], SwinJSCC [2], and MambaJSCC [5] provide the codec context for our experiments. The separately trained direct-codec references in Section IV locate the reported quality relative to other architectures.

Other methods adapt reconstruction within an inference procedure. ISEC iteratively corrects latent errors using a learned source prior [6]. DiffCom conditions generative reconstruction on channel observations and includes blind inference of transmission characteristics [7]. GibbsDDRM estimates unknown forward-model parameters during diffusion inference [8]. Our decision occurs before such iterative processing: it selects which disturbance treatment to invoke. We study the usefulness and cost of that choice, without proposing a new diffusion prior or posterior inference algorithm.

### III. SYSTEM MODEL AND RECEIVER CHOICE

#### A. An Interfering Latent Transmission

Let  $\mathbf{u} \in [0, 1]^d$  be the desired image, and let the frozen encoder produce  $\mathbf{x} \in \mathbb{C}^n$ . A separate transmitter produces interfering symbols  $\mathbf{z} \in \mathbb{C}^n$ . Each transmitter normalizes its frame to unit average complex symbol power. The additive channel is

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{w}_b, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad a_b \geq 0, \quad (1)$$

where  $b$  indexes  $B$  contiguous blocks, with  $L_b$  complex symbols per block. The channel bandwidth ratio is  $n/d$ . The receiver knows  $N_0$ , the normalization convention, and a chosen block partition, but not  $a_b$ , the interference profile,  $\mathbf{z}$ , or  $\mathbf{u}$ . The test channel has unit desired-channel gain. Fading, channel estimation, and synchronization are outside this experiment.

For a prescribed nominal SINR  $\gamma$ , the simulator sets average nominal interference power to  $10^{-\gamma/10} - N_0$ . The instantaneous interference energy also depends on the realized block energies of  $\mathbf{z}$ ; nominal SINR must therefore not be confused with measured per-frame SINR.

The explicit model associates two learned priors with (1):

$$p(\mathbf{x}, \mathbf{z} \mid \mathbf{y}, \mathbf{a}) \propto p_s(\mathbf{x}) p_z(\mathbf{z}) \times \exp \left[ -\frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|^2 \right]. \quad (2)$$

This equation describes the model; our finite-step sampler is an approximation, not an exact posterior draw or a proven MAP optimizer. Without priors, the change  $\mathbf{x}'_b = \mathbf{x}_b + a_b \delta_b$ ,  $\mathbf{z}'_b = \mathbf{z}_b - \delta_b$  leaves the mixture unchanged. Power constraints remove some ambiguity but do not generally identify both latents. The Gaussian model instead uses

$$p_G(\mathbf{y} \mid \mathbf{x}; \nu) = \mathcal{CN}(\mathbf{y}; \mathbf{x}, \nu \mathbf{I}), \quad \nu \geq N_0. \quad (3)$$

It sacrifices interference structure to avoid explicit component estimation.

#### B. Two Frozen Candidate Receivers

Both candidates process the same received frame and use the same desired-signal prior and decoder. They differ in how they represent the disturbance. For Gaussian reception, estimate the disturbance from raw received symbols:

$$e_b = \frac{\|\mathbf{y}_b\|^2}{L_b}, \quad \bar{e} = \frac{1}{n} \sum_b L_b e_b, \quad (4)$$

$$\hat{P} = [\bar{e} - 1 - N_0]_+, \quad \hat{\nu} = N_0 + \hat{P}.$$

With unit source power and vanishing signal–interference cross expectation, unclipped excess energy estimates mean interference energy. Finite blocks, latent correlations, and clipping prevent treating it as an exact amplitude estimate. In particular, local desired-signal energy can resemble interference heterogeneity.

Let  $r(\mathbf{y}) \in \mathbb{R}^{2n}$  stack real and imaginary components. Given the training diffusion schedule  $\bar{\alpha}_t$ , the Gaussian receiver G starts at the nearest noise level satisfying

$$\frac{1 - \bar{\alpha}_{t_0}}{\bar{\alpha}_{t_0}} \simeq \frac{\hat{\nu}}{2}, \quad \mathbf{v}_{t_0} = \sqrt{\bar{\alpha}_{t_0}} r(\mathbf{y}). \quad (5)$$

The factor 1/2 converts complex noise power to variance per real coordinate. For decreasing respaced indices, a noise predictor  $\epsilon_s$  gives

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{v}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_s(\mathbf{v}_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (6)$$

$$\mathbf{v}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \epsilon_s(\mathbf{v}_t, t). \quad (7)$$

At most 40 predictor evaluations are used, with fewer when the matched initial index is below 39. A common normalization and the frozen decoder produce the reconstructed image.

The structured receiver S initializes  $a_b^{(0)} = \sqrt{[e_b - 1 - N_0]_+}$  and performs joint diffusion inference with signal and interference predictors. During late corrector updates it obtains denoised, frame-normalized estimates  $\tilde{\mathbf{x}}$  and  $\tilde{\mathbf{z}}$ , then updates

$$a_b^+ = \left[ \frac{\Re\{\tilde{\mathbf{z}}_b^H (\mathbf{y}_b - \tilde{\mathbf{x}}_b)\}}{\|\tilde{\mathbf{z}}_b\|^2} \right]_+. \quad (8)$$

An empty denominator retains the previous amplitude. This is a nonnegative least-squares update conditional on point estimates, not a correction for their uncertainty. Calibration begins when the signal coefficient  $\alpha(t) \geq 0.5$ . Estimated block SINR updates the inherited guidance coefficients by clipped linear interpolation of the original AWGN table. We use the existing 40-step, order-three joint sampler: 40 signal and 40 interference evaluations. The initialization-only variant keeps amplitudes fixed and supplies an ablation of calibration.

#### C. Selection Before Generative Inference

The selector augments  $\hat{P}$  with normalized block-energy variation

$$H = \frac{B^{-1} \sum_{b=1}^B (e_b - \bar{e})^2}{\max(\bar{e}^2, \varepsilon)}. \quad (9)$$

Here  $\varepsilon > 0$  prevents division by zero, and all receiver blocks have equal length.  $H$  measures energy variation, not the interference distribution. For  $\theta = (\tau_\ell, \tau_h, \eta)$ , define

$$\pi_\theta(\mathbf{y}) = \begin{cases} G, & \hat{P} \leq \tau_\ell, H \leq \eta, \\ G, & \hat{P} \leq \tau_h, H > \eta, \\ S, & \text{otherwise.} \end{cases} \quad (10)$$

We also test the reversed assignment of G and S in (10); the Gaussian branch is not assumed to belong to the low-power region. The energy-only family sets  $\tau_\ell = \tau_h$ . Either family can reduce to a constant receiver; heterogeneity need not be useful.

We fit thresholds to mean per-image PSNR on development data. Eight folds keep all channel conditions belonging to one image together. Each fold is predicted using thresholds fitted on the other seven, then the family with the highest

out-of-fold mean is refitted on all development images. Ties prefer lower component inference cost and then the simpler family. If neither family improves on the better fixed receiver in development validation, that fixed receiver is retained. No threshold is adjusted using confirmation or pressure results. At reception, (10) is computed first and only its chosen branch is executed. Selection adds  $O(n)$  energy calculations and no diffusion evaluations.

For analysis, let  $q_j(\mathbf{u}, \mathbf{y})$  be a candidate's PSNR. The unavailable per-frame oracle has quality  $\max(q_G, q_S)$ . The exact identity

$$\max(q_G, q_S) - q_{\pi_\theta} = |q_G - q_S| \mathbf{1}\{\pi_\theta \neq \arg \max_j q_j\} \quad (11)$$

shows that complementary receivers are necessary but not sufficient: mistakes on large-gap frames matter most. This is an accounting identity, not a generalization bound. Mean PSNR is the fitting objective; neither mean MSE, LPIPS nor harmful-tail probability is implicitly optimized.

#### IV. EXPERIMENTAL EVALUATION

##### A. Data, Receivers, and Evaluation Protocol

The desired images are  $128 \times 128$  center crops from CelebA [9]. CIFAR-10 interference images are cropped at  $32 \times 32$  and nearest-neighbor upsampled to the encoder input size. The original training sets contain 162,770 desired and 50,000 interference images. The inherited codec uses three Swin stages with widths (128, 192, 256) and depths (2, 2, 6); its 8-channel  $16 \times 16$  real latent corresponds to 1,024 complex symbols and CBR 1/48. Five frozen checkpoints contain the desired encoder/decoder, interference encoder, and two DiT predictors, each with 28 blocks, width 1,152, 16 heads and patch size two. The original frozen-study training schedule uses 100 interference-codec epochs, 40 desired-codec epochs, and 50/20 epochs for the signal/interference predictors. The selector study adds no neural retraining. The two direct codec anchors are separately retrained for 40 epochs at SNR 20 dB with 1,024 complex channel uses (CBR 1/48); each also trains a matched CIFAR-10 interference encoder. Best checkpoints minimize held-out validation MSE; training uses one seed. DeepJSCC uses a pinned third-party PyTorch reproduction, while MambaJSCC uses its official repository. We also test transfer using repository-released weights without fine-tuning. Public MambaJSCC retains CBR 1/48 but differs in training domain, channel and resolution; the available DeepJSCC weight uses CBR 0.16256. These protocol-mismatched diagnostics use the same public encoder for both image streams and are reported separately from the retrained references.

Development uses 64 validation image pairs and seed 1700. Confirmation reserves 256 previously unevaluated test pairs with seeds 2700 and 2701. SNR is 20 dB; nominal SINR is  $-7, -4, 0, 4, 7, 10$  dB. Sixteen blocks of 64 complex symbols follow stationary, alternating (0.25, 1.75), or burst (0, 2) relative-power patterns. Each pair is evaluated under all 18 conditions. The confirmation set therefore has 9,216 received frames but only 256 image-pair clusters. Confirmation starts

only after policy fitting ends. File-hash audits exclude overlap with development, pressure images and the first 200 test pairs used in earlier project evaluations. The independence claim is at image-pair level, not at person-identity level.

We compare direct decoding; the same Gaussian port using only  $N_0$ ; the blind-total-disturbance Gaussian receiver G; initialization-only joint sampling; calibrated joint sampling S; and the online selector. A same-gate control decodes directly when  $\hat{P} \leq 0.15$  and otherwise uses S. Its output quality is reconstructed exactly from paired branches; its reported cost excludes gate overhead. Development additionally examines the previous coupled-feedback receiver and a Gaussian port given nominal total disturbance power. The nominal-power receiver is a privileged development diagnostic, not a guaranteed upper bound. DeepJSCC and MambaJSCC direct decoding are reported as absolute anchors using their own desired and interference codewords under the same channel grid. The diffusion priors are specific to the SwinJSCC latent distribution; attaching them to another codec would require new prior training. Reproducing complete CDDM/ICDM training and decoder adaptation remains outside this receiver study, so published-system competitiveness remains unestablished.

The prespecified power thresholds are 0, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, 8 and  $10^6$ ; heterogeneity thresholds are  $\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$ . Including both assignment directions, the families contain 26 and 2,028 candidate rules. All SwinJSCC receiver variants see identical received tensors. The two direct anchors use separately encoded, unit-power codewords at the same budget and reuse the image pairs, conditions, and channel seeds. Channel randomness is separate from sampler randomness; both CPU and GPU sampler states are reset per frame. Measurements use float32, batch one, TF32 disabled, four CPU threads, and one RTX 4090. Synchronized receiver time includes selection, sampling and decoding, but excludes encoding, image-quality evaluation and one-time model loading. Signal and interference model calls are counted separately.

We report arithmetic means of per-image PSNR and MSE, LPIPS-VGG v0.1 [10], and four-scale MS-SSIM [11], always clipping RGB to  $[0, 1]$ . Four scales fit the image resolution; these values are not compared with five-scale scores. Paired 95% intervals resample image-pair clusters 5,000 times while retaining both seeds and all conditions. Harm means a PSNR loss exceeding 0.5 dB relative to a specified comparator. The resampling unit is the image pair rather than individual frames.

##### B. Frozen Rule and Receiver Comparison

Image-group validation yields 22.002 dB for energy-only selection and 22.239 dB for the two-feature family. The retained energy and heterogeneity rule uses  $\tau_\ell = 8$ ,  $\tau_h = 0.5$  and  $\eta = 0.2$ , with Gaussian reception below the applicable power threshold.

In the low-heterogeneity development region (810 frames), the largest observed excess-energy estimate is 5.241; the fitted threshold 8 therefore lies beyond observed support and acts as an always-Gaussian rule in that region. It is not an

TABLE I

INDEPENDENT CONFIRMATION. QUALITY IS AVERAGED PER IMAGE AND FRAME. NFE COUNTS BOTH PRIORS; LATENCY IS SYNCHRONIZED BATCH-ONE RECEIVER TIME. THE SAME-GATE CONTROL’S LATENCY IS A COMPONENT ESTIMATE. DEEPJSCC AND MAMBAJSCC ROWS ARE MATCHED-RETRAINED DIRECT-CODEC ANCHORS; ALL REMAINING ROWS SHARE THE FROZEN SWINJSCC CODEC.

| Receiver         | PSNR  | MSE $\times 10^3$ | LPIPS | MS-SSIM | NFE  | ms   |
|------------------|-------|-------------------|-------|---------|------|------|
| SwinJSCC direct  | 18.67 | 29.14             | 0.440 | 0.697   | 0.0  | 8    |
| DeepJSCC direct  | 17.39 | 24.71             | 0.575 | 0.684   | 0.0  | 1    |
| MambaJSCC direct | 16.92 | 44.13             | 0.463 | 0.613   | 0.0  | 10   |
| Gaussian (N0)    | 18.83 | 28.70             | 0.438 | 0.700   | 18.0 | 225  |
| Blind Gaussian   | 20.84 | 24.90             | 0.378 | 0.782   | 40.0 | 488  |
| Init. joint      | 19.61 | 20.15             | 0.363 | 0.793   | 80.0 | 1110 |
| Calibrated joint | 20.26 | 18.49             | 0.362 | 0.802   | 80.0 | 1180 |
| Joint + gate     | 20.45 | 18.43             | 0.361 | 0.801   | 65.9 | 974  |
| Selector         | 21.59 | 16.45             | 0.336 | 0.832   | 52.2 | 701  |

identified physical transition, and the family comparison does not establish that either feature is individually necessary. The earlier coupled-feedback receiver obtains 20.477 dB, versus 20.605 dB for calibrated joint sampling and 20.835 dB for its same-gate control. The privileged nominal-power Gaussian diagnostic obtains 21.591 dB. These development comparisons motivate the receiver decision but are not independent confirmation. Table I and Fig. 1 report the independent experiment. The selector changes mean PSNR by +1.334 dB relative to calibrated joint sampling (95% interval [1.265, 1.404]), +0.750 dB relative to blind Gaussian denoising (95% interval [0.698, 0.806]), and +1.140 dB relative to the same-gate control (95% interval [1.075, 1.205]). Both fixed-receiver comparisons have positive paired intervals, supporting a mean-PSNR benefit within the matched confirmation population. Estimating total disturbance changes the Gaussian receiver from 18.829 to 20.843 dB, so the thermal-noise-only port is insufficient as the sole Gaussian baseline.

The fixed receivers are complementary across interference profiles. For stationary interference, Gaussian denoising reaches 21.310 dB against 19.282 dB for joint sampling. Under burst interference, joint sampling reaches 21.317 dB against 20.481 dB for Gaussian denoising. The selector reaches 21.317, 21.285, and 22.177 dB for stationary, alternating, and burst interference, respectively. Thus its gain is consistent with using different receivers for different observations; the energy statistic alone does not establish a physical explanation of the interference.

Separately matched direct-codec anchors attain 17.394 dB for DeepJSCC and 16.923 dB for MambaJSCC; these use independently trained latent spaces and provide external codec references. One training seed and a fixed 40-epoch budget do not establish convergence or a general architecture ranking. Without fine-tuning, public DeepJSCC (ImageNet/SNR 19), MambaJSCC (CLIC2021/AWGN 10), and MambaJSCC (DIV2K/Rayleigh) weights yield 13.755, 17.020, and 16.650 dB, respectively. Their protocol mismatches preclude a controlled claim about retraining benefit.

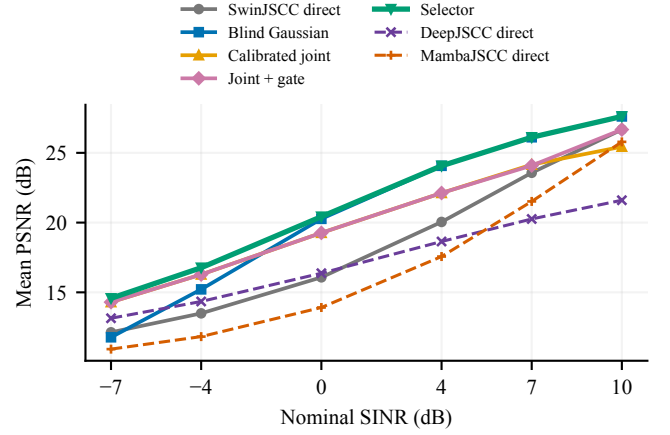


Fig. 1. Independent confirmation by nominal SINR, averaging all three interference profiles. Solid curves compare receivers in the frozen SwinJSCC latent space; dashed curves are separately retrained direct-codec anchors.

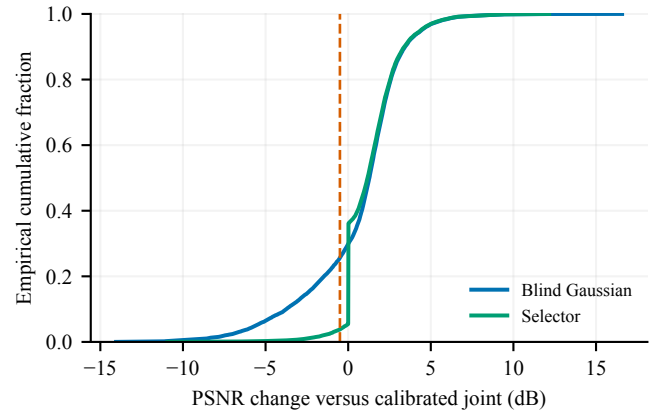


Fig. 2. Paired PSNR change on independent confirmation relative to calibrated joint sampling. Each point in the empirical CDF represents one received frame; uncertainty calculations cluster by image pair. The vertical line marks the predefined harmful-loss threshold.

### C. Inference Cost and Harmful Decisions

The selector chooses Gaussian reception on 69.5% of confirmation frames. Its average cost is 52.2 predictor evaluations and 701 ms, versus 80 evaluations and 1180 ms for joint sampling. Against that receiver, 3.79% of frames lose more than 0.5 dB, and the fifth percentile of the paired change is -0.127 dB (Fig. 2). The unavailable per-frame oracle reaches 21.760 dB; selection leaves 0.167 dB mean oracle regret. Mean MSE changes from 0.01849 to 0.01645, LPIPS from 0.3618 to 0.3362, and MS-SSIM from 0.8015 to 0.8318. These secondary metrics and the harmful tail must accompany the average PSNR comparison.

### D. Block Mismatch and No Interference

Pressure tests use 64 further test pairs and seed 3700, with nominal SINR -4, 4, 10 dB. One profile shifts burst boundaries by half a receiver block; another draws segment lengths

TABLE II  
PRESSURE TESTS. MEAN PSNR FOR EACH FAMILY. THE  
NO-INTERFERENCE ROW HAS SNR 20 dB; THE OTHER ROWS AVERAGE  
NOMINAL SINR  $-4, 4, 10$  dB.

| Profile         | Direct | Gaussian | Joint | Select |
|-----------------|--------|----------|-------|--------|
| Shifted blocks  | 20.83  | 22.99    | 21.09 | 23.00  |
| Random lengths  | 20.64  | 23.42    | 22.74 | 23.98  |
| No interference | 31.37  | 31.37    | 27.73 | 31.37  |

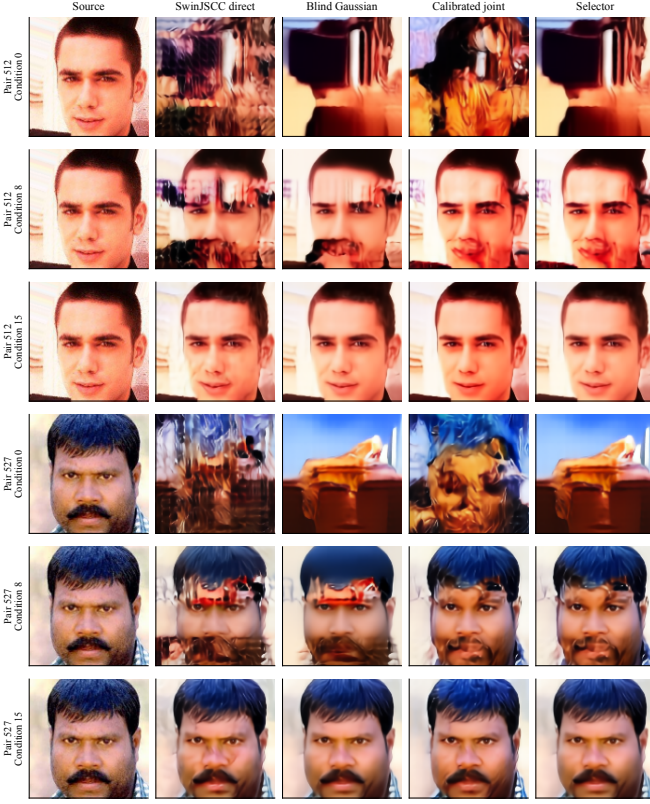


Fig. 3. All six qualitative examples reserved before confirmation: two image pairs crossed with conditions 0, 8 and 15. The gallery is exhaustive for the prespecified set and is not selected from observed reconstruction quality.

uniformly from 16 to 128 symbols and normalizes the binary on/off amplitude pattern. A separate SNR-20 condition has no interference. The receiver keeps its original block partition and frozen rule. These families are reported separately from matched confirmation. Table II shows a  $+0.010$  dB selector change against Gaussian reception for shifted blocks and  $+0.563$  dB for random lengths. With no interference, selection matches Gaussian reception and avoids the joint receiver's loss. These are diagnostic populations; no rule is refitted or result pooled with matched confirmation.

## V. DISCUSSION AND CONCLUSION

On independent image pairs, the frozen selector improves mean PSNR over both blind Gaussian denoising and calibrated joint sampling. It also improves on the same-gate control. The same-gate result shows that the gain cannot be explained

solely by bypassing joint inference. The gain is supported for the evaluated source, interference population and inference budget. It is not a guarantee for individual frames: the harmful tail persists, and block-boundary mismatch nearly removes the advantage over Gaussian denoising. Received-energy heterogeneity can also originate from the desired signal, so the branch label should be interpreted as a reconstruction decision rather than identification of the interferer.

PSNR and MSE aggregate the same pixel errors differently; LPIPS and MS-SSIM add perceptual evidence but do not measure success in a driving, recognition or control task. Applying this receiver decision to vehicular communication requires evaluation with task-specific losses, fading, channel estimation and synchronization. The present results establish a narrower point: selecting between existing disturbance models can improve average reconstruction while reducing the cost of always invoking joint inference.

## REFERENCES

- [1] E. Boursoulatz, D. B. Kurka, and D. Gündüz, "Deep joint source-channel coding for wireless image transmission," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 3, pp. 567–579, 2019.
- [2] K. Yang, S. Wang, J. Dai, X. Qin, K. Niu, and P. Zhang, "SwinJSCC: Taming swin transformer for deep joint source-channel coding," *IEEE Trans. Cogn. Commun. Netw.*, vol. 11, no. 1, pp. 90–104, 2025.
- [3] T. Wu, Z. Chen, D. He, L. Qian, Y. Xu, M. Tao, and W. Zhang, "CDDM: Channel denoising diffusion models for wireless semantic communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11 168–11 183, Sep. 2024.
- [4] T. Wu, Z. Chen, D. He, F. Yang, M. Tao, X. Xu, W. Zhang, and P. Zhang, "ICDM: Interference cancellation diffusion models for wireless semantic communications," *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 2528–2543, 2026.
- [5] T. Wu, Z. Chen, M. Tao, Y. Sun, X. Xu, W. Zhang, and P. Zhang, "MambaJSCC: Adaptive deep joint source-channel coding with generalized state space model," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 9264–9279, 2026.
- [6] C. Lee, X. Hu, and H.-S. Kim, "Deep joint source-channel coding with iterative source error correction," in *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, F. Ruiz, J. Dy, and J.-W. van de Meent, Eds., vol. 206. PMLR, 25–27 Apr 2023, pp. 3879–3902. [Online]. Available: <https://proceedings.mlr.press/v206/lee23c.html>
- [7] S. Wang, J. Dai, K. Tan, X. Qin, K. Niu, and P. Zhang, "Diffcom: Channel received signal is a natural condition to guide diffusion posterior sampling," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 7, pp. 2651–2666, 2025. [Online]. Available: <http://dx.doi.org/10.1109/JSAC.2025.3559158>
- [8] N. Murata, K. Saito, C.-H. Lai, Y. Takida, T. Uesaka, Y. Mitsufuji, and S. Ermon, "GibbsDDRM: A partially collapsed gibbs sampler for solving blind inverse problems with denoising diffusion restoration," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 25 501–25 522.
- [9] Z. Liu, P. Luo, X. Wang, and X. Tang, "Deep learning face attributes in the wild," in *2015 IEEE International Conference on Computer Vision (ICCV)*. IEEE, Dec. 2015, pp. 3730–3738. [Online]. Available: <http://dx.doi.org/10.1109/ICCV.2015.425>
- [10] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, 2018, pp. 586–595. [Online]. Available: <http://dx.doi.org/10.1109/CVPR.2018.00068>
- [11] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale structural similarity for image quality assessment," in *The Thirty-Seventh Asilomar Conference on Signals, Systems & Computers*, 2003, ser. ACSSC-03. IEEE, 2003, pp. 1398–1402. [Online]. Available: <http://dx.doi.org/10.1109/ACSSC.2003.1292216>